MAT 221 - CALCULUS I
Lesson 21 - 22 - Sequence and Series

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Sequence

- A sequence of real numbers is a function whose domain is the set of positive integers. The values taken by the function are called terms of the sequence.
- It is an ordered pattern of numbers defined by a function whose inputs are strictly positive or negative integers and the output in the form of real values.

Notation

- The terms of a sequence are denoted by $\{a_n\}$.
- The function f that takes in the terms is given by $a_n = f(n)$.

Examples of Sequences

Sequence	Terms of the Sequence
$\{a_n\} = \{3n-2\}_{n=1}^{\infty}$	$1, 4, 7, 10, ...$(an A.P sequence)
$\{a_n\} = \{\frac{1}{n}\}_{n=1}^{\infty}$	$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, ...$(harmonic sequence)
$\{a_n\} = \{4(\frac{1}{2})^{n-1}\}_{n=1}^{\infty}$	$4, 2, 1, \frac{1}{2}, ...$(a G.P sequence)
$\{a_n\} = \{\frac{1}{n!}\}_{n=1}^{\infty}$	$1, \frac{1}{2}, \frac{1}{6}, \frac{1}{24}, ..$
$\{a_n\} = \{\frac{n(n+1)}{2}\}$	$1, 3, 6, 10, 15...$(Sequence of triangular numbers)

Boundedness

(i) The sequence $\{a_n\}$ is **bounded above** if there is a number M_1 such that $\{a_n\} \leqslant M_1$ for every positive integer n.

(ii) The sequence $\{a_n\}$ is **bounded below** if there is a number M_2 such that $M_2 \leqslant \{a_n\}$ for every positive integer n.

(iii) It is bounded if there is a number $M > 0$ such that $|a_n| \leqslant M$ for every positive integer n.

(iv) If the sequence is not bounded, it is called **unbounded**.

Remark

- The numbers M_1, M_2 and M are called an upper bound, a lower bound and a bound for $\{a_n\}$.
- If $\{a_n\}$ is bounded above and below, then it is bounded. Simply set $M = max\{|M_1, |M_2|\}$.

Examples

(a) The sequence $\{(-1)^n\}$ has the upper bound 1, the lower bound -1, and the bound 1.

(b) The sequence $\{\cos n\}$ has the upper bound of 1, the lower bound of -1, and the bound 1 since $-1 \leqslant \cos n \leqslant 1$ for every n.

(c) The sequence $\{3^n\}$ is bounded below by 3 for $n \geqslant 1$ but has no upper bound and so is unbounded.

(d) The sequence $\{(-1)^n 2^n\}$ is bounded neither below or above.

Finite limit of a sequence

A sequence has the limit L if for every $\epsilon > 0$ there exist an integer $N > 0$ such that if $n \geqslant N$, then $|a_n - L| < \epsilon$. We write

$$\lim_{n\to\infty} a_n = L. \tag{1}$$

This implies that $a_n \to L$ as n increases without bound.

Infinite Limit of a Sequence

The sequence $\{a_n\}$ has the limit ∞ if for every positive number M there is an integer $N > 0$ such that if $n > N$, then $a_n > M$. In this case, we write

$$\lim_{n\to\infty} a_n = \infty. \tag{2}$$

Intuitively, $\lim_{n\to\infty} a_n = \infty$ means that n increases without bound a_n increases without bound. $\lim_{n\to\infty} a_n = -\infty$ is defined in a similar way.

Convergence and Divergence of a Sequence

If the limit in (1) exists and if L is finite, we say that the sequence converges or is convergent. Otherwise, we say that the sequence diverges or is divergent.

Example Determine whether the following sequences are convergent or divergent.

(a) $a_n = \{\frac{1}{3}\}^n$.

(b) $a_n = 1 - 2n$.

Solutions

(a) We have,

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{1}{3}\right)^n = 0.$$

Hence, the sequence is convergent.

(b) We have,

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (1 - 2n) = -\infty.$$

Hence, the sequence is divergent.

Theorem 1

If the sequence $\{a_n\}$ is convergent, then it is bounded.

The theorem implies that every convergent sequence is bounded and every unbounded sequence is divergent. However, we should note that not every bounded sequence is convergent.

Monotonicity

- The sequence $\{a_n\}$ is monotone increasing if $a_{n+1} \geqslant a_n$ for every $n \geqslant 1$.
- The sequence $\{a_n\}$ is monotone decreasing if $a_{n+1} \leqslant a_n$ for every $n \geqslant 1$.
- The sequence $\{a_n\}$ is monotonic if it either monotone increasing or monotone decreasing.
- The sequence $\{a_n\}$ is strictly increasing if $a_{n+1} > a_n$ for every $n \geqslant 1$.
- The sequence $\{a_n\}$ is strictly decreasing if $a_{n+1} < a_n$ for every $n \geqslant 1$.
- The sequence $\{a_n\}$ is strictly monotonic if it is either strictly increasing or strictly decreasing.

Example

Determine whether the sequences are increasing, decreasing or not monotonic.

(a) $a_n = \frac{1}{3^n}$,

(b) $a_n = \frac{2n}{3n+2}$

Solution

(a) The sequence $a_n = \frac{1}{3^n}$, is strictly decreasing since $a_{n+1} < a_n$, i.e, $\frac{1}{3^n} > \frac{1}{3^{n+1}}$, for every n

(b) The sequence $\frac{3n}{2n+1}$ is strictly increasing because, using calculus; $f(x) = \frac{2x}{3x+2}, f''(x) = \frac{4}{(3x+2)^2} > 0$. hence f is increasing for $x \neq \frac{-2}{3}$. But $a_{n+1} = f(x+1) > f(n) = a_n$. So $a_n = \frac{2n}{3n+2}$, is strictly increasing.

Series

A series is a sum of all terms of a sequence. It is usually denoted by;

$$\sum_{n=1}^{\infty} a_n.$$

Examples of Series

Arithmetic Sequence and Series

- An arithmetic progression (A.P) is a linear sequence with a starting term a and common difference d is defined by

$$T_n = a + (n-1)d. \tag{3}$$

Where T_n is the term and n is the position of the term in the sequence. The sequence is given by;

$$a, a+d, a+2d, a+4d, ..., a+(n-1)d, ...$$

- **Arithmetic Series**: It is the sum of all terms of an A. P up to n^{th} term;

$$S_n = a + a + d + a + 2d + a + 3d + ... + a + (n-1)d.$$

The formula for an arithmetic series is,

$$S_n = \frac{n}{2}[2a + (n-1)d]. \tag{4}$$

Example

The 6th and 11th terms of an A.P are -16 and -31. Calculate the sum of the first 30 terms of the A.P.

Solution

$$a + 5d = -16$$

$$a + 10d = -31$$

Solving we get $a = -3$ and $d = 2$. Therefore

$$S_{30} = \frac{30}{2}[2(-3) + 29 \times 2] = .$$

Exercise

(a) Find 19th term of an A.P, given that the 8th term is 13 and the sum of the first 10 terms is 55.

(b)

(c)

Geometric Progression (G.P) and Series

- It is a sequence where each succeeding term is produced by multiplying each preceding term by a fixed number, which is called a common ratio.
- The formula is given by;

$$T_n = ar^{n-1}.$$

  - a is the first term.
  - r is the common ratio.
  - n is the position of the term.
- G.P sequence: $a, ar, ar^2, ar^3, ..., ar^{n-1}, ...$
- **G.P Series**: It is the sum of a G.P up to the nth term.

$$S_n = a + ar + ar^2 + ar^3 + ... + ar^{n-1}$$

Given by the formula,

$$S_n = \frac{a(1 - r^n)}{1 - r}. \tag{5}$$

Example The 5th term of a G.P sequence is 162 and the 8th term is 4374. Find the sum of the first 10 terms.

Solution

$$T_5 = ar^4 = 162.$$

$$T_8 = ar^7 = 4374$$

Solving by dividing:

$$\frac{ar^7}{ar^4} = \frac{4374}{162} = 27$$

$r^3 = 27 \rightarrow r = 3$ and $a = 2$.
Sum of the first 10 terms: $S_{10} = \frac{2(1-3^{10})}{1-3} = 59048.$

Exercise

(a)

(b)

(c)